

Foundations of the $\mathbf{MMK}$ Category

Definition of Objects and Morphisms in $\mathbf{MMK}$

Objects of $\mathbf{MMK}$ (Metric-Measure-Kernel Systems). An object X in the category $\mathbf{MMK}$ is a quadruple $X = (|X|, d, \mu, K)$ consisting of:

- A **state space** $|X|$, which is equipped with a metric $d: |X| \times |X| \rightarrow \mathbb{R}_{\geq 0}$ (typically a complete separable metric, making $|X|$ a Polish space ¹). The metric induces a Borel σ -algebra $\mathcal{B}(|X|)$.
- A **probability measure** μ on $(|X|, \mathcal{B}(|X|))$, so $\mu(|X|) = 1$ ².
- A **Markov kernel** $K: |X| \times \mathcal{B}(|X|) \rightarrow [0, 1]$, which assigns to each state $x \in |X|$ a probability measure $K(x, \cdot)$ on $|X|$ (interpreted as the one-step transition distribution from x). Measurability in x is assumed (each map $x \mapsto K(x, A)$ is measurable for fixed A) ³. Often K is assumed *Feller*, meaning it interacts continuously with bounded continuous functions ¹, though this technical condition will not affect the categorical logic development.

Morphisms of $\mathbf{MMK}$ (Markov Coarse-Grainings). A morphism $F: X \rightarrow Y$ between two objects $(|X|, d_X, \mu_X, K_X)$ and $(|Y|, d_Y, \mu_Y, K_Y)$ is a measurable function $F: |X| \rightarrow |Y|$ satisfying two preservation properties:

1. **Measure preservation:** $F_{\#} \mu_X = \mu_Y$. That is, the pushforward of the measure μ_X by F equals μ_Y . Equivalently, for every measurable set $B \subseteq |Y|$, $\mu_X(F^{-1}(B)) = \mu_Y(B)$ ⁴. Intuitively, F redistributes the states of X into those of Y without changing the overall probability distribution.
2. **Kernel intertwining:** $F_{\#} K_X = K_Y$, meaning $K_Y(F(x), B) = K_X(x, F^{-1}(B))$ for all measurable $B \subseteq |Y|$, at least for μ_X -almost every $x \in |X|$ ⁵ ⁴. In other words, if $x \in X$ maps to $y = F(x) \in Y$, then the distribution of one-step transitions from x (under K_X) corresponds exactly to the distribution of transitions from y (under K_Y) when mapped back along F . This *exact intertwining* condition ensures F preserves the stochastic dynamics: the probability of transitioning into any set $B \subseteq |Y|$ from state x equals the probability (from x) of transitioning into the preimage of B in X , which then maps into B . Equivalently, $K_Y(F(x), B) = (F_{\#} K_X)(x, B)$ where $(F_{\#} K_X)(x, B) := K_X(x, F^{-1}(B))$ ⁴.

A morphism $F: X \rightarrow Y$ is thus a **Markov coarse-graining**: it *lumps* or aggregates the states of X into states of Y in a way that preserves both the static distribution and the one-step transition probabilities ⁶. The intuition is that Y is a simplified or abstracted system obtained by identifying certain states of X via F , without losing probabilistic information. If $F(x) = F(x')$, the two states x, x' of X are treated as the same abstract state in Y , so F forgets the distinction between them, but it does so only if their behavior (distribution μ_X and transitions K_X) can be merged consistently.

Examples: (a) The identity map $\mathrm{id}_X: X \rightarrow X$ trivially satisfies $(\mathrm{id}_X)_\mu_X = \mu_X$ and $(\mathrm{id}_X)_{K_X} = K_X$, thus is a morphism. (b) If Y is a coarse-grained version of X where some states of X are amalgamated, the canonical projection $F: X \rightarrow Y$ (mapping each state to its equivalence class) can be an $\mathbb{MMK}$ -morphism *if and only if* the partition respects the measure and kernel as above. For instance, grouping a set of states $E \subseteq X$ into one state in Y requires that $K_X(x, E) = 1$ for all $x \in E$ (they form a closed communicating class under K_X) and $\mu_X(E)$ becomes the probability of that new state in μ_Y .

Isomorphisms: An $\mathbb{MMK}$ -morphism $F: X \rightarrow Y$ is an **isomorphism** if it has an inverse morphism $G: Y \rightarrow X$ (with $G \circ F = \mathrm{id}_X$ and $F \circ G = \mathrm{id}_Y$). Concretely, F is an iso if there exists a bijective measurable map G such that F and G are measure-preserving and intertwine the kernels both ways. In practice, this means F is bijective *up to null sets* (both F and G may ignore sets of measure zero) and it carries μ_X onto μ_Y and K_X onto K_Y (and vice versa). Thus, X and Y are essentially the *same* metric-measure-kernel system, differing at most on a set of states of measure zero. In particular, $|X|$ and $|Y|$ must be isomorphic as measure spaces, and F must carry the stochastic dynamics of X exactly to that of Y . Any such isomorphism preserves all “experiences” of states: if $F(x) = y$, then for every sequence of transitions or measurable property, x and y have matching outcomes. Later we will see that if $\mathrm{Pred}(X)$ separates points of X (which it does under mild conditions), then an isomorphism $F: X \cong Y$ can also be characterized as a map that reflects all predicates: $\varphi(x) = \psi$ implies there is a corresponding φ' on Y with $\varphi'(F(x)) = \psi$, etc. Intuitively, an isomorphism in $\mathbb{MMK}$ is a *relabeling* of states that leaves all metric, measure, and kernel structure intact.

Subobjects: A **subobject** of X in $\mathbb{MMK}$ can be thought of as a subsystem represented by a subset of states of $|X|$ with inherited structure. In classical category terms, a subobject is an equivalence class of monomorphisms into X . Here a monomorphism would be a morphism $m: S \rightarrow X$ that is injective (one-to-one) in the sense that if two morphisms $h_1, h_2: Z \rightarrow S$ satisfy $m \circ h_1 = m \circ h_2$, then $h_1 = h_2$. In $\mathbb{MMK}$, any *measurable subset* $E \subseteq X$ (with appropriate conditions) can yield a subobject: we take $S = (E, d|_E, \mu_E, K_E)$ where $d|_E$ is the restricted metric on E , μ_E is the **conditional measure** on E (the probability measure obtained by restricting μ_X to E and renormalizing so that $\mu_E(E) = 1$ ⁷), and K_E is the **conditional kernel** on E (obtained by restricting K_X to transitions within E and renormalizing so that for each $x \in E$, $K_E(x, \cdot)$ is a probability distribution on E) ⁷. The inclusion map $\iota: E \hookrightarrow X$ given by $\iota(x) = x$ is then a candidate morphism $S \rightarrow X$. Indeed, by construction ι_μ_E is the original μ_X restricted to E and then extended by zero outside E ; since we renormalized, ι_μ_E is not equal to μ_X unless $\mu_X(E) = 1$. However, ι does satisfy the kernel condition: for $x \in E$ and measurable $B \subseteq X$, $(\iota_{K_E})(x, B) = K_E(x, \iota^{-1}(B)) = K_E(x, B) = K_X(x, B)$ (because $B \subseteq E$), and for $B \subseteq X$ not contained in E , $B \cap E$ is the relevant part since $K_E(x, E) = 1$ for $x \in E$. In effect, ι is an $\mathbb{MMK}$ -morphism if E is K_X -invariant ($K_X(x, E) = 1$ for $x \in E$) and $\mu_X(E) = 1$ (so no renormalization needed). In general, proper subsets will not strictly satisfy measure preservation, so the notion of subobject in $\mathbb{MMK}$ is subtle: one often works with invariant full-measure subsets. For example, the equalizer of two morphisms (see below) provides a natural example of a subobject of X . One can summarize that subobjects correspond to measurable embeddings* $S \rightarrow X$ where S is a subsystem of X (possibly requiring invariance conditions for strict adherence to $\mathbb{MMK}$ rules). In later developments, certain reflective subcategories (like those defined by closure operators) will yield subobjects via embeddings that are monomorphisms in a weakened sense (embedding as a dense or full subset) ⁸.

Basic Category Properties

Proposition 1: $\mathbf{MMK}$ is a well-defined category. The composition of $\mathbf{MMK}$ -morphisms is again a morphism, and identity morphisms exist, so the usual category axioms hold ⁹. Specifically:

- *Identities:* For each object $X=(|X|,d,\mu,K)$, the identity map $\mathrm{id}_X: |X| \rightarrow |X|$ (sending each state to itself) is an $\mathbf{MMK}$ morphism. Indeed, trivially $\mathrm{id}_K = K$, since pulling back along the identity does nothing. The identity morphism acts as the neutral element for composition. $\mu = \mu$ and $\mathrm{id}_{X\setminus\#}$
- *Composition:* Given morphisms $F: X \rightarrow Y$ and $G: Y \rightarrow Z$, their composition $G \circ F: X \rightarrow Z$ is also a morphism ⁹. This follows from the functorial properties of pushforward: $(G \circ F)_\# \mu_X = G_\#(F_\# \mu_X) = G_\# \mu_Y = \mu_Z$ (so measure preservation holds) ¹⁰, and similarly for the kernel $(G \circ F)_* K_X = G_*(F_* K_X) = G_* K_Y = K_Z$ ¹⁰. Thus $\mathbf{MMK}$ is closed under composition, and associativity is inherited from the associativity of function composition. (The equalities here hold in the sense of measures; any necessary measurable selection or almost-everywhere considerations can be handled because if F and G each satisfy their intertwining conditions μ -a.e., then $(G \circ F)_\#$ will satisfy the needed condition μ_X -a.e. by Fubini's theorem or a two-step application of the a.e. equalities.)

Proof Sketch: The verification is straightforward. Identity is immediate. For composition, note that $F_\# \mu_X = \mu_Y$ and $G_\# \mu_Y = \mu_Z$ imply $(G \circ F)_\# \mu_X = \mu_Z$ by the transitivity of pushforwards ¹⁰. Likewise $K_Z(F(x), G^{-1}(B))$ equals $K_Y(F(x), B')$ for $B' = G^{-1}(B)$ (since G intertwines K_Y and K_Z), which equals $K_X(x, F^{-1}(B')) = K_X(x, (G \circ F)^{-1}(B))$ (since F intertwines K_X and K_Y). These equalities hold for μ_X -almost every x , establishing that $G \circ F$ satisfies the kernel condition and hence is a morphism. Identity and associativity laws then hold by construction ⁹.

Isomorphisms and Equivalences: If $F: X \rightarrow Y$ is an isomorphism in $\mathbf{MMK}$, as noted above, it means F is essentially bijective and preserves all structure. We often consider two objects *equivalent* (in the sense of probability and dynamics) if there exists an isomorphism between them. This captures the idea that $\mathbf{MMK}$ does not distinguish systems that differ only by renaming states or ignoring null sets. In measure-theoretic contexts, one often identifies morphisms that agree μ -almost everywhere; here we implicitly do the same, treating morphisms that differ only on μ_X -null sets as essentially identical (since all conditions are a.e.). Thus, in many arguments we may assume morphisms and isomorphisms are defined up to null sets, ensuring, for example, that every isomorphism has a genuine inverse on the supports of the measures.

Monomorphisms and Subobjects: A morphism $m: S \rightarrow X$ in $\mathbf{MMK}$ is a **monomorphism** if it is injective in the categorical sense. Concretely, one expects m to be (almost) injective on states. In fact, if m is measure-preserving and (a.e.) injective as a function, then m is a monomorphism (since any two morphisms $h_1, h_2: Z \rightarrow S$ with $m \circ h_1 = m \circ h_2$ would imply $h_1(z) = h_2(z)$ whenever $m(h_1(z))$ —which equals $m(h_2(z))$ —is in the support of μ_X , forcing $h_1 = h_2$ a.e.). Intuitively, a monomorphism $m: S \rightarrow X$ identifies S as a sub-system of X . As discussed, non-trivial subobjects require care: if S covers only part of X 's measure, m cannot strictly preserve μ_X . However, one typically considers *full-support* subobjects (those hitting all of μ_X or at least all x of interest) or works in an extended setting of sub-probability measures. In later sections, our **closure**

operators $\text{clos}\{a\}, \text{clos}\{g\}, \text{clos}\{s\}, \text{clos}\{\text{ell}\}$ will produce subobjects of a given X that *are* reflective and satisfy certain invariance, making them well-behaved substructures (e.g. dense embeddings or invariant subsets) ¹¹.

Finite Limits in $\mathbf{MMK}$

A desirable feature of $\mathbf{MMK}$ is that it admits **finite limits**, which include in particular a **terminal object**, **binary products**, and **equalizers** of parallel morphisms. These constructions allow one to perform certain universal operations within the category and will be useful for defining logical structures (e.g. pullbacks for quantification) and certain quotient systems.

- **Terminal Object:** A terminal object I in a category is one that has a unique morphism from any object. In $\mathbf{MMK}$, the terminal object is the **one-point system** $I = (\{ \}, d \equiv 0, \delta, \text{id})$ consisting of a single state $\{ \}$ (distance 0 to itself), the Dirac measure δ concentrated at that state, and the identity Markov kernel $K_{(\cdot), \{ \}} = 1$ (the only possible kernel on a one-point space) ¹². For any object $X = (|X|, d, \mu, K)$, there is a unique morphism $! : X \rightarrow I$, given by mapping every $x \in |X|$ to $\{ \}$ in I . This map is trivially measurable and satisfies $(!)^* \mu_X = \delta$ (since δ assigns probability 1 to $\{ \}$ and μ_X of the preimage of $\{ \}$ is $\mu_X(|X|) = 1$) and $(!)^* K_X = \text{id}$ (since for the unique $y = \{ \}$ in I , $K(y, \cdot)$ is trivial and $K_X(x, (\cdot)^{-1}(\cdot)) = K_X(x, |X|)$ which equals 1 by probability kernel property). Thus I is indeed terminal. Intuitively, I represents a collapsed system with no information (everything is one state); morphisms to I forget everything about the input system except the fact that "it exists". Having a terminal object means we can talk about **points** or **state selections** of a system via morphisms from I into it (though I being initial would correspond to that; here terminal gives a way to collapse systems).

- **Binary Products:** Given two objects $X = (|X|, d_X, \mu_X, K_X)$ and $Y = (|Y|, d_Y, \mu_Y, K_Y)$ in $\mathbf{MMK}$, their categorical product is an object $X \times Y$ together with projection morphisms $\pi_X : X \times Y \rightarrow X$ and $\pi_Y : X \times Y \rightarrow Y$, satisfying the universal property of the product. We construct $X \times Y$ concretely as follows ¹³ ¹⁴:

- The state space is the Cartesian product $|X \times Y| = |X| \times |Y|$.
- The metric $d_{X \times Y}$ is taken as a product metric, for example

$$d_{X \times Y}((x_1, y_1), (x_2, y_2)) := \sqrt{d_X(x_1, x_2)^2 + d_Y(y_1, y_2)^2},$$

which makes $|X| \times |Y|$ a metric space (indeed Polish if X and Y are Polish). This choice ensures that topologically, $X \times Y$ is the product space and is separable and complete if X and Y are. (Any equivalent metric would do; this ℓ^2 -style metric is a standard choice.)

- The measure on $X \times Y$ is the **product measure** $\mu_X \otimes \mu_Y$, which is the unique probability measure on the product σ -algebra such that $(\mu_X \otimes \mu_Y)(A \times B) = \mu_X(A) \cdot \mu_Y(B)$ for measurable $A \subseteq |X|$, $B \subseteq |Y|$. Since μ_X and μ_Y are probability measures, $\mu_X \otimes \mu_Y$ is also a probability measure on $|X| \times |Y|$ ¹³.

- The Markov kernel on $X \times Y$ is defined as the **independent product kernel** $K_{X \times Y} = K_X \otimes K_Y$. This kernel acts as if X and Y evolve independently: for a product state $(x, y) \in |X| \times |Y|$ and measurable rectangle $A \times B \subseteq |X| \times |Y|$, we set

$$(K_X \otimes K_Y)((x, y), A \times B) := K_X(x, A) \cdot K_Y(y, B),$$

and extend this to all measurable subsets by standard measure-theoretic constructions (via Carathéodory's extension theorem) ¹³. In words, from state (x, y) , the next-step distribution factorizes: the X -component jumps according to $K_X(x, \cdot)$ independently of the Y -component, which jumps according to $K_Y(y, \cdot)$. This makes $(X \times Y)$ essentially the system where X and Y run in parallel with no interaction.

Now define the **projection morphisms** $\pi_X: X \times Y \rightarrow X$ by $\pi_X(x, y) = x$ and $\pi_Y: X \times Y \rightarrow Y$ by $\pi_Y(x, y) = y$. These maps are clearly measurable (continuous, even) and by construction satisfy pushforward conditions: $(\pi_X)_*(\mu_{X \times Y}) = \mu_X$ and $(\pi_Y)_*(\mu_{X \times Y}) = \mu_Y$ (marginals of the product measure are the original measures). They also intertwine the kernels: for instance, $K_X(\pi_X(x, y), A) = K_X(x, A)$, and

$$(\pi_X)_*(K_X \otimes K_Y)((x, y), A) = (K_X \otimes K_Y)((x, y), A \times |Y|) = K_X(x, A) K_Y(y, |Y|) = K_X(x, A),$$

since $K_Y(y, |Y|) = 1$. Thus π_X pushes the product kernel to K_X , and similarly π_Y does for K_Y ¹⁵. It is straightforward to verify the universal property: given any object Z and morphisms $f: Z \rightarrow X$, $g: Z \rightarrow Y$, there is a unique MMK -morphism $h: Z \rightarrow X \times Y$ such that $\pi_X \circ h = f$ and $\pi_Y \circ h = g$. This h is just $h(z) = (f(z), g(z))$ on the underlying sets. One must check h preserves measure and kernel: indeed, $h_*\mu_Z = (f_*\mu_Z) \otimes (g_*\mu_Z) = \mu_X \otimes \mu_Y$ because $f_*\mu_Z = \mu_X$ and $g_*\mu_Z = \mu_Y$ (here we use independence or Fubini to argue $(f, g)_*\mu_Z = f_*\mu_Z \otimes g_*\mu_Z$ for the probability measures). And similarly $h_*K_Z = K_X \otimes K_Y$ if $f_*K_Z = K_X$ and $g_*K_Z = K_Y$. Thus $X \times Y$ with π_X, π_Y is the categorical product. In particular, MMK is a symmetric monoidal category under $\otimes$, with unit object I (the one-point space) acting as the monoidal unit (since $X \otimes I \cong X$ clearly). The monoidal structure formalizes the idea of combining independent systems: $\otimes$ allows us to consider composite systems consisting of two subsystems that do not interact. This will be important when reasoning about multiple systems or factors (e.g. combining a geometric and a logical component, etc.). Note:* The monoidal product $\otimes$ here is just the categorical product; we use the tensor symbol to emphasize the interpretation of an independent joint system. It is symmetric ($X \otimes Y \cong Y \otimes X$ naturally) and associative up to isomorphism, as usual for categorical products.

- **Equalizers:** Given two parallel morphisms $F_1, F_2: X \rightarrow Y$ in MMK , their **equalizer** is a morphism $m: E \rightarrow X$ such that $F_1 \circ m = F_2 \circ m$, and which is universal with this property (any other object E' mapping into X that equalizes F_1, F_2 factors uniquely through m). Intuitively, E picks out the part of X on which F_1 and F_2 agree. We construct E as:

$$|E| := \{x \in |X| : F_1(x) = F_2(x)\},$$

the equalizer set of the two underlying functions ¹⁶. We endow E with the subspace metric $d_X|_{|E|}$ and define μ_E as the normalized restriction of μ_X to E ⁷. That is, $\mu_E(A) := \frac{\mu_X(A)}{\mu_X(E)}$ for measurable $A \subseteq E$ (assuming $\mu_X(E) > 0$; if $\mu_X(E) = 0$, the equalizer is trivial or not considered an object since it would not carry a probability measure). For the

kernel, we define K_E as the conditional kernel on E : for $x \in E$, $K_E(x, \cdot)$ is the probability law of one K_X -step conditioned on remaining in E . Formally, for a measurable subset $B \subseteq E$,

$$K_E(x, B) := \frac{K_X(x, B)}{K_X(x, E)},$$

provided $K_X(x, E) > 0$ (and $K_X(x, E)$ will equal 1 for $x \in E$ if leaving E has zero probability from x ; otherwise we consider that if x can leave E , we condition on the event of staying in E in one step). If $K_X(x, E) = 0$ for some $x \in E$, then x cannot reach E in one step, which is problematic for K_E ; in practice, one typically restricts attention to cases where E is K_X -invariant (as noted above) or else one works with a sub-stochastic kernel. Under the assumption that F_1, F_2 are equal on E in a strong sense (so that if $x \in E$, then any step from x stays in E with probability 1—this holds if $F_1(x) = F_2(x)$ implies $K_X(x, F_1^{-1}(y)) = 1$ for that y , etc.), K_E is well-defined and $K_E(x, E) = 1$. We then have an inclusion morphism $\iota: E \hookrightarrow X$ defined by $\iota(x) = x$. This ι **equalizes** F_1 and F_2 because if $x \in E$, $F_1(\iota(x)) = F_1(x) = F_2(x) = F_2(\iota(x))$ by definition of E . Moreover, ι is an MMK -morphism: $\iota_* \mu_E$ is μ_X restricted to E (extended by zero outside E), which is not the same as μ_X on all of X unless $\mu_X(E) = 1$, but since ι is not surjective onto X , it doesn't need to equalize those measures on X outside E . What matters is that ι is monomorphic and any map from another object Z that equalizes F_1, F_2 must factor through E . The universal property is satisfied: if $H: Z \rightarrow X$ is any morphism with $F_1 \circ H = F_2 \circ H$, then $\text{im}(H) \subseteq E$ (at least μ_Z -almost surely), so H factors (a.e.) through $\iota: E \rightarrow X$. We thus say E with ι is the equalizer.

In simpler terms, ignoring measure-zero subtleties, E is the subset of X where F_1 and F_2 literally agree. The measure μ_E is μ_X conditioned on that subset, and the kernel K_E is the original kernel conditioned not to leave that subset. If F_1 and F_2 differ on a set of positive μ_X -measure, then $\mu_X(E) < 1$ and ι is not measure-preserving onto X (because it misses some mass); nonetheless, E as an object captures the essence of the equalizer. In practice, many interesting equalizers involve F_1, F_2 that agree on a high-probability region or on an invariant set. In extreme cases, if F_1 and F_2 are substantially different, the equalizer might be a null set (no nontrivial E), indicating no meaningful subobject where they coincide.

Proposition 2: Finite limits exist in MMK . In summary, MMK has a terminal object, binary products, and equalizers (hence all finite limits exist) ¹⁷ ¹² .

Proof Sketch: Terminal object I was described above. Binary products $X \times Y$ are constructed as above; verification of the universal property is standard by checking the defining equalities on measures and kernels (using independence and Fubini to handle composition with arbitrary morphisms into $X \times Y$) ¹⁵ . For equalizers, one must check that if $H: Z \rightarrow X$ equalizes F_1, F_2 , then essentially $H(Z) \subseteq E$ and a factorization $Z \rightarrow E \rightarrow X$ exists as a morphism. Under mild conditions (e.g. $H_* \mu_Z$ is supported in E and K_Z transitions remain in $H^{-1}(E)$), such a factorization can be defined and is unique. A rigorous construction might consider the image measure $H_* \mu_Z$ restricted to E and the appropriate kernel, ensuring the factor map is measure-preserving and intertwining, but the intuitive argument suffices: any state z that H sends to X outside of E would make $F_1(H(z)) \neq F_2(H(z))$, contradicting the equalizing property. Therefore $H(z)$ lies in E for all relevant z , and one can define $\bar{H}: Z \rightarrow E$ by $\bar{H}(z) = H(z)$ (which lands in E). This $\bar{H}$ is a well-defined morphism $Z \rightarrow E$ and $\iota \circ \bar{H} = H$. Uniqueness is clear (since ι is injective). Thus the equalizer is universal ⁷ . $\square$

Remark: The existence of finite limits in $\mathcal{MMK}$ means it is a *finitely complete* category. This is important for the development of an internal logic, as it guarantees that pullbacks (in particular, pullbacks of F along itself to get equalizers, etc.) exist. We will see next how having finite limits enables the formulation of logical quantifiers via an indexed predicate functor. Note that while finite limits exist, **finite colimits** in $\mathcal{MMK}$ are more delicate. For example, a categorical coproduct (disjoint union) of two probability spaces does not naturally carry a single probability measure unless one component is given zero weight or we drop the probability normalization. Likewise, coequalizers (which would correspond to quotienting and “merging” states of an object) need not exist in $\mathcal{MMK}$ in general unless certain conditions hold (the $\mathcal{Q}$ functor described elsewhere provides a *particular* quotient under specific circumstances ^{6 18}). However, the **coarse-graining functor** Q (mentioned in later sections) will serve as a kind of “best possible” coequalizer for certain diagrams by producing a quotient object, albeit not every equivalence relation comes from an $\mathcal{MMK}$ morphism. In summary, $\mathcal{MMK}$ is not cocomplete, but it has enough limits and structure for internal logic and reflective subcategories.

Internal Logic and Predicate Hyperdoctrine

One of the powerful features of $\mathcal{MMK}$ is that it supports an **internal logic** – a way to interpret logical predicates and quantifiers *within* the category. Each object of $\mathcal{MMK}$ carries a σ -complete lattice of predicates, and each morphism induces logical quantifiers (existential and universal) acting as adjoints to pullback. This structure makes $\mathcal{MMK}$ an **indexed category of predicates** or **hyperdoctrine** in the sense of F. W. Lawvere, enabling reasoning about properties of states in a functorial, compositional manner ¹⁹. We define this logical structure formally:

Definition 3: Predicates on an object. For an object $X = (|X|, d, \mu, k)$ in $\mathcal{MMK}$, define $\text{Pred}(X)$ to be the set of all $\mathcal{B}(|X|)$ -measurable functions $\varphi: |X| \rightarrow [0,1]$. An element of $\text{Pred}(X)$ is thought of as a **predicate** (or property) of states in X that may hold to a certain degree in $[0,1]$. We interpret the value $\varphi(x) \in [0,1]$ as the **truth value** of predicate φ at state x . Here 1 represents “completely true” and 0 “completely false,” with intermediate values representing fuzzy or probabilistic truth. (In particular, characteristic functions of measurable sets are $\{0,1\}$ -valued predicates identifying crisp subsets of $|X|$.) We order predicates **pointwise**: for $\varphi, \psi \in \text{Pred}(X)$, we write $\varphi \leq \psi$ if $\varphi(x) \leq \psi(x)$ for all $x \in |X|$. This gives $\text{Pred}(X)$ the structure of a complete lattice ²⁰: any set of predicates has a supremum (least upper bound) and infimum (greatest lower bound) defined pointwise by sup/inf of the values in $[0,1]$. In particular, we have the **logical connectives** defined pointwise for $\varphi, \psi \in \text{Pred}(X)$:

- (*Conjunction*) $(\varphi \wedge \psi)(x) := \min\{\varphi(x), \psi(x)\}$, representing the predicate “ φ AND ψ ,” true to the lesser degree of truth of φ or ψ at x ²¹.
- (*Disjunction*) $(\varphi \vee \psi)(x) := \max\{\varphi(x), \psi(x)\}$, representing “ φ OR ψ ,” true to the greater degree that either holds at x ²¹.
- (*Negation*) $(\neg \varphi)(x) := 1 - \varphi(x)$, representing logical NOT, with truth value the complement in $[0,1]$ ²².

These operations make each $\text{Pred}(X)$ a *complete lattice* (indeed a σ -complete Boolean algebra if we restricted to $\{0,1\}$ -valued predicates, but here we allow $[0,1]$ -valued “fuzzy” predicates, which forms an algebra of continuous truth values). The top element (always true) is the constant predicate $1_X(x) \equiv 1$ and the bottom (false) is $0_X(x) \equiv 0$. We can also interpret *implication* between predicates: we say φ **entails** ψ (written $\varphi \Vdash \psi$) iff $\varphi(x) \leq \psi(x)$ for all x , i.e. φ

$\leq \psi$ in the lattice order. In logical terms, this means at every state, whenever φ holds to some degree, ψ holds at least to that degree – in particular, if $\varphi(x)=1$ (completely true) then we must have $\psi(x)=1$ as well; more generally a weaker condition cannot be more true than a stronger condition at any point. Entailment in this fuzzy setting thus generalizes subset inclusion (for characteristic predicates) to the quantitative realm.

Pullback of predicates (reindexing). Any $\mathbf{MMK}$ -morphism $F: X \rightarrow Y$ induces a mapping of predicates from Y to X by pre-composition. Formally, define

$$F^* : \mathbf{Pred}(Y) \rightarrow \mathbf{Pred}(X)$$

by $(F^*\psi)(x) := \psi(F(x))$ for $\psi: Y \rightarrow [0,1]$ ²³. This is just the usual pullback (inverse image) of functions along F . Since F is measurable and ψ is measurable, $F^*\psi$ is measurable on X . The pullback F^* preserves all the logical structure: it is a lattice homomorphism (infima and suprema are preserved) because $(F^*(\bigvee_i \psi_i))(x) = \bigvee_i \psi_i(F(x)) = \bigvee_i (F^*\psi_i)(x)$, etc. In particular $F^*(\psi_1 \wedge \psi_2) = F^*\psi_1 \wedge F^*\psi_2$, and $F^*(\neg \psi) = \neg (F^*\psi)$. We thus obtain an **indexed category of predicates**, i.e. a functor

$$\mathbf{Pred} : \mathbf{MMK}^{op} \rightarrow \mathbf{CLat}$$

from the opposite of $\mathbf{MMK}$ to the category of complete lattices (and lattice homomorphisms) ²³. In simple terms, each object has a predicate lattice, and each morphism $F: X \rightarrow Y$ gives a *predicate transformer* $F^*: \mathbf{Pred}(Y) \rightarrow \mathbf{Pred}(X)$ sending each predicate on Y to its pullback (or pre-image) on X . This corresponds to interpreting a property of Y in the context of X via F : for $A \subseteq Y$, $F^*(\mathbf{1}_A) = \mathbf{1}_{F^{-1}(A)}$, and for a general fuzzy predicate ψ , $F^*\psi$ evaluates ψ at the image state $F(x)$. $\mathbf{1}_A$ (the characteristic of $F^{-1}(A)$)

Pullback is contravariant (reverses arrows), reflecting that if X maps to Y , then predicates of Y pull back to predicates of X . This structure $(\mathbf{Pred}, F^*)$ is the first ingredient in an internal logic: it assigns to each system X a logic (predicates on X) and to each mapping $F: X \rightarrow Y$ a substitution operation (F^*) that allows one to compare or relate predicates between X and Y .

Adjoint quantifiers: A hallmark of logical hyperdoctrines is the presence of **left and right adjoints** to pullback, which correspond to **existential** and **universal quantification** along a morphism. In our probabilistic setting, these quantifiers will be realized by taking essential supremum and infimum over fibers. The key fact, guaranteed by the structure of $\mathbf{MMK}$, is that for each $\mathbf{MMK}$ -morphism $F: X \rightarrow Y$, the monotone map $F^*: \mathbf{Pred}(Y) \rightarrow \mathbf{Pred}(X)$ has both a left adjoint $\exists_F$ and a right adjoint $\forall_F$ in the lattice order ²⁴.

- The **existential quantifier** (left adjoint) $\exists_F: \mathbf{Pred}(X) \rightarrow \mathbf{Pred}(Y)$ is defined for each $\varphi \in \mathbf{Pred}(X)$ as

$$(\exists_F \varphi)(y) := \operatorname{ess\,sup}_{x \in F^{-1}(y)} \varphi(x).$$

Here $F^{-1}(y)$ denotes the fiber $\{x \in X : F(x)=y\}$, and $\operatorname{ess\,sup}$ is the essential supremum with respect to μ_X conditional on $F(x)=y$ ²⁵. Intuitively, $(\exists_F \varphi)(y)$ gives the maximum truth value of φ among all states x in X that map to y , disregarding a

μ_X -null set of exceptional x if needed. In other words, it asserts that “there exists an x in the fiber over y such that $\varphi(x)$ is true,” to the extent of the highest φ -truth in that fiber. If any x in $F^{-1}(y)$ makes $\varphi(x)$ nearly 1, then $(\exists_F \varphi)(y)$ will be 1; if $\varphi(x)=0$ for all x in the fiber, then $(\exists_F \varphi)(y)=0$; and intermediate values reflect the supremal truth found in the fiber. This generalizes the logical $\exists$ (existential) quantifier to $[0,1]$ -valued predicates: it is like a “fuzzy or” over the fiber.

- The **universal quantifier** (right adjoint) $\forall_F: \text{Pred}(X) \rightarrow \text{Pred}(Y)$ is defined dually as

$$(\forall_F \varphi)(y) := \text{ess inf}_{x \in F^{-1}(y)} \varphi(x).$$

So $(\forall_F \varphi)(y)$ is the **minimum truth value** of φ on the fiber over y (again up to null exceptions) ²⁵. It reflects that “for all x in the fiber, $\varphi(x)$ holds,” to the extent of the worst-case (smallest) truth of φ in that fiber. If any x in the fiber falsifies φ (making it 0), then $(\forall_F \varphi)(y)=0$; only if $\varphi(x)$ is 1 for μ_X -almost every x in the fiber will $(\forall_F \varphi)(y)=1$. This corresponds to a $\forall$ (universal) quantifier in logic, taking a “fuzzy and” over the fiber.

These are indeed well-defined measurable predicates on Y ²⁵. The measurability of $\exists_F \varphi$ and $\forall_F \varphi$ as functions $|Y| \rightarrow [0,1]$ relies on a Fubini or parameterization argument: since $|X|$, $|Y|$ are standard Borel spaces and F is measurable, one can ensure the essential sup/inf over fibers varies measurably with y (this uses a property known as *Suslin's theorem* or existence of regular conditional distributions in some cases, but intuitively it's fine because we can consider a version of φ saturated on each fiber) ²⁶.

By construction, $\exists_F$ and $\forall_F$ are adjoint to F^* in the sense of order adjunctions:

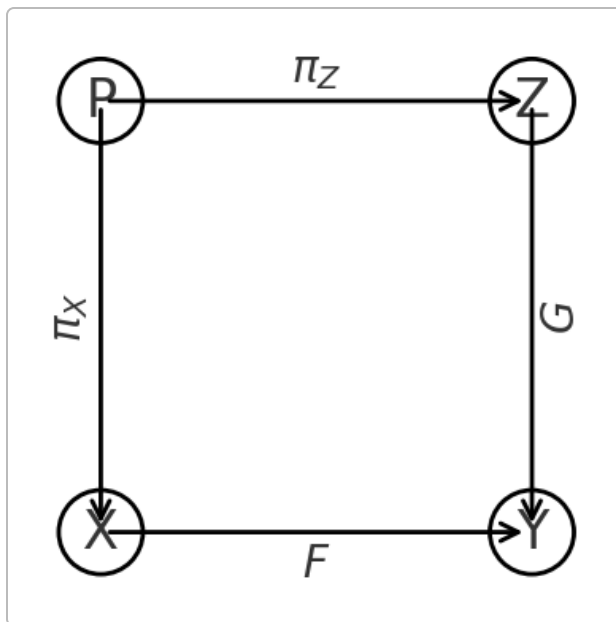
$$\exists_F(\varphi) \leq \psi \iff \varphi \leq F^*(\psi), \quad \text{and} \quad F^*(\psi) \leq \varphi \iff \psi \leq \forall_F(\varphi),$$

for all $\varphi \in \text{Pred}(X)$, $\psi \in \text{Pred}(Y)$ ²⁷. These adjunction laws* capture the essence of existential and universal quantifiers as left and right adjoints respectively: $\exists_F$ is the least predicate on Y whose pullback exceeds φ , and $\forall_F$ is the greatest predicate on Y whose pullback is below φ .

In more familiar logical terms: $\exists_F \varphi$ on Y says “ $\varphi(x)$ is true for some x with $F(x)=y$,” and $\forall_F \varphi$ says “ $\varphi(x)$ is true for all x with $F(x)=y$.” The adjunction $\exists_F \varphi \leq \psi \iff \varphi \leq F^* \psi$ can be read as: “Existence of an x such that $\varphi(x)$ implies $\psi(F(x))$ if and only if for each x , $\varphi(x)$ implies $\psi(F(x))$,” which is a form of logical correctness for $\exists$; similarly for $\forall$.

These quantifiers satisfy expected logical properties: - **Monotonicity**: $\exists_F$ preserves join ($\vee$) and $\forall_F$ preserves meet ($\wedge$). For example, $\exists_F(\varphi_1 \vee \varphi_2) = (\exists_F \varphi_1) \vee (\exists_F \varphi_2)$, because taking suprema over fibers commutes with taking suprema of the functions (sup over a union of sets = sup of sups) ²⁵. Dually, $\forall_F(\varphi_1 \wedge \varphi_2) = (\forall_F \varphi_1) \wedge (\forall_F \varphi_2)$, since inf over fiber of a pointwise min is min of infs. They also obviously preserve 0 and 1: $\exists_F(0)=0$, $\exists_F(1)=1$, and $\forall_F(0)=0$, $\forall_F(1)=1$. - **Beck-**

Chevalley condition: Quantifiers commute properly with pullback along *pullback squares*. Concretely, given a commutative pullback diagram in $\mathbf{MMK}$ as shown below:



Pullback square in $\mathbf{MMK}$ with $P = X \times_Y Z$.

Suppose

$$\begin{array}{ccc} P & \xrightarrow{\pi_Z} & Z \\ \downarrow \pi_X & & \downarrow G \\ X & \xrightarrow{F} & Y \end{array}$$

is a pullback square (meaning $P = \{(x,z) \in X \times Z : F(x) = G(z)\}$, with π_X, π_Z the coordinate projections, which are $\mathbf{MMK}$ -morphisms)²⁸. Then for any predicate $\Phi \in \text{Pred}(X)$ we have the

Beck-Chevalley equalities:

$$\exists_{\pi_Z}(\pi_X^* \Phi) = G^*(\exists_F \Phi), \quad \forall_{\pi_Z}(\pi_X^* \Phi) = G^*(\forall_F \Phi).$$

²⁹ These state that whether we pull Φ back along π_X and then push forward along π_Z , or first push Φ forward along F and then pull back along G , we get the same result, and similarly for the universal quantifier. Intuitively, this is a **change-of-variable** or **parametricity** rule: if Z is another object mapping to Y , and $P = X \times_Y Z$ is the fiber product, then $\exists_F$ followed by reindexing along G equals reindexing along π_X followed by $\exists_{\pi_Z}$. This is analogous to the statement in logic that exchanging the order of summation/integration or logical operations in a commuting situation yields the same outcome. The proof uses the fact that fibers are in bijection: for a given $z \in Z$ with $G(z) = y$, the fiber $\pi_Z^{-1}(z)$ corresponds exactly to the fiber $F^{-1}(y)$ via π_X . So taking suprema or infima over $\pi_X^{-1}(y)$ and then interpreting in Z is the same as pulling $\exists_F \Phi$ or $\forall_F \Phi$ back to that z ³⁰.

- **Frobenius reciprocity:** The quantifiers distribute over finite logical operations in a specific way. For any $\varphi \in \text{Pred}(X)$ and $\psi \in \text{Pred}(Y)$, we have:

$$\exists_F(\varphi \wedge F^*\psi) = (\exists_F\varphi) \wedge \psi, \quad \forall_F(\varphi \vee F^*\psi) = (\forall_F\varphi) \vee \psi.$$

³¹ These are sometimes called the Frobenius laws. The first says that if we have a property $\varphi(x)$ of x and an external condition $\psi(F(x))$ about the image of x , then “ $\exists$ an x with $\varphi(x)$ and $F(x)$ satisfying ψ ” is equivalent to “($\exists$ an x with $\varphi(x)$) and ψ holds for that y .” Formally, $\exists_F(\varphi \wedge F^*\psi)(y) = \sup_{x: F(x)=y} \min\{\varphi(x), \psi(y)\}$. But since $\psi(y)$ is constant for a fixed y , this equals $\min\{\psi(y), \sup_{x: F(x)=y} \varphi(x)\} = (\exists_F\varphi)(y) \wedge \psi(y)$ ³². The second law is dual: “for all x with $\varphi(x)$ or (some condition on y) holds” is equivalent to “(for all x with $\varphi(x)$) or that condition on y holds”. It is proved by a similar pointwise argument using infimum and maximum ³². These properties ensure that quantification interacts well with the logical connectives, a crucial aspect of internal logic (it implies, for example, that the quantifiers are complete homomorphisms* of the lattice in a certain fibered sense, preserving finite meets or joins up to these equalities).

Given these structures, $(\text{Pred}, \wedge, \vee, \neg, F^*, \exists_F, \forall_F)$ forms a *predicate hyperdoctrine* (in the sense of Lambek & Scott ³³): an indexed lattice with each fiber a (complete) logic, each arrow giving adjoint quantifiers. In categorical logic terms, MMK behaves akin to a many-valued logic with truth values in $[0,1]$. In fact, each $\text{Pred}(X)$ is an example of a $[0,1]$ -valued Boolean algebra (sometimes called a logic algebra or tribe in probability logic), and the functorial assignment $X \mapsto \text{Pred}(X)$, $F \mapsto (F^* \dashv \exists_F \dashv \forall_F)$ endows MMK with an **internal many-valued (or fuzzy) logic**.

Some observations about this internal logic: - If we restrict predicates to only 0 or 1 values (characteristic functions of measurable sets), the logic reduces to the usual Boolean logic of measurable subsets (subobject logic). $\exists_F$ on a characteristic predicate corresponds to the image of a set under F (up to null sets), and $\forall_F$ corresponds to the condition that the entire fiber is within a set. - The presence of real-valued (in $[0,1]$) predicates allows degrees of truth, which is appropriate for probabilistic semantics. This is reminiscent of **fuzzy logic** or **probability logic** where one deals with truth values in $[0,1]$. Here $\wedge = \min, \vee = \max, \neg(x) = 1-x$ corresponds to the standard fuzzy logic (Gödel logic) operations. The adjoints we have are then a form of *quantitative existential/universal* that are common in fuzzy description logics or in reasoning about probability (they are essentially the Lowe–Gaifman linear preorders in probabilistic logic, realized as sup and inf). - The adjointness conditions ensure a **sound semantics**: for instance, $\varphi \leq F^*\psi$ iff $\exists_F \varphi \leq \psi$ means that if for every x with property $\varphi(x)$, the image $F(x)$ satisfies ψ , then the existence of an x with $\varphi(x)$ implies that ψ holds in the image. This matches our intuition about existential quantification being logically sound. Similarly, $\forall_F \varphi \leq \psi$ being right adjoint says if pulling back ψ is always $\leq \varphi$, then ψ itself is $\leq \forall_F \varphi$, i.e. if whenever $F(x)=y$ we have $\psi(y)$ only as true as $\varphi(x)$, then $\psi(y)$ can be no greater than the infimum of $\varphi(x)$ on that fiber.

Internal Semantics: Probability-Valued Logic and Stochastic Reasoning

We now interpret the above machinery as a **semantic framework** for reasoning about MMK systems. Each object X carries a **probability-valued logic** $\text{Pred}(X)$, where each predicate $\varphi(x) \in [0,1]$ can

be thought of as the probability (or degree) to which some property holds at state x . The truth values are in the interval $[0,1]$, which one can consider as the **truth value object** in this many-valued logic (in fact $[0,1]$ itself is the unit interval lattice). The connectives $(\wedge, \vee, \neg)$ allow the usual combinations of properties. For example, if $\varphi(x)$ is the fuzzy event “ x satisfies property P ” and $\psi(x)$ is “ x satisfies Q ,” then $(\varphi \wedge \psi)(x)$ is “ x satisfies both P and Q ” with truth value the minimum of $\varphi(x)$ and $\psi(x)$. Negation gives the complement property.

Truth values and entailment: A predicate $\varphi \in \text{Pred}(X)$ is **totally true** in state x if $\varphi(x)=1$ and **totally false** if $\varphi(x)=0$. Usually, one might say “ x satisfies φ ” (notation $x \models \varphi$) if $\varphi(x)=1$ for μ_X -almost all x (i.e., the property holds almost surely in X). Entailment $\varphi \rightarrow \psi$ (meaning $\varphi \leq \psi$ pointwise) corresponds to the statement “for every state, if φ holds (to any degree) then ψ holds at least as much.” In particular, if $\varphi(x)=1$ then necessarily $\psi(x)=1$ in every state x . Thus $\varphi \rightarrow \psi$ captures the idea that ψ is true whenever φ is true – a typical semantic entailment. In the special case of $\{0,1\}$ -predicates, $\varphi \rightarrow \psi$ means $\mathbf{1}_A \leq \mathbf{1}_B$, which is equivalent to $A \subseteq B$ (modulo null sets). So entailment generalizes subset inclusion to fuzzy predicates.

States as models: Each individual state $x \in |X|$ can be seen as a *model* (or possible world) that assigns truth values to all predicates in $\text{Pred}(X)$. The structure $(|X|, \text{Pred}(X))$ is akin to a many-valued model in logic. Because $\text{Pred}(X)$ is very rich (all measurable functions), it separates states under mild conditions: typically, for any two distinct states $x_1 \neq x_2$, there exists some predicate φ with $\varphi(x_1) \neq \varphi(x_2)$ (for instance, if the metric is separable, one can use a sequence of continuous predicates or approximate Dirac functions to distinguish points ³⁴). Therefore, the internal logic is *adequate* to distinguish states (except perhaps on a set of measure zero or in pathological cases). This has an important consequence: if two states x_1, x_2 yield $\varphi(x_1) = \varphi(x_2)$ for **all** predicates $\varphi \in \text{Pred}(X)$, then from the logic’s perspective they are **indiscernible** (logically equivalent) ³⁵. In fact, one can define a logical equivalence relation $\sim_\ell$ on $|X|$ by

$$x_1 \sim_\ell x_2 \iff \forall \varphi \in \text{Pred}(X), \varphi(x_1) = \varphi(x_2).$$

This relation identifies states that have exactly the same “experiences” or observable properties in terms of all predicates ³⁶. Because $\text{Pred}(X)$ is so expressive (including essentially all measurable properties), typically $\sim_\ell$ is just equality (or equality mod null sets if the σ -algebra cannot separate some pathological points). However, in some cases where X has redundant or unobservable distinctions, $\sim_\ell$ might be a nontrivial equivalence that collapses those distinctions. The **logical closure** functor $\text{clos}_\ell(X)$ mentioned in later sections essentially factors X by $\sim_\ell$ ³⁷, yielding a quotient object where states are exactly equivalence classes of $\sim_\ell$. In that quotient, any two distinct states can be distinguished by some predicate (by construction), meaning $\text{clos}_\ell(X)$ is a *fully abstract* or *separated* model with respect to the internal logic. The metric on $\text{clos}_\ell(X)$ can be induced by logical distances (like supremum norm of predicate differences ³⁸), which often coincides with the original metric if it was already separating measure-theoretic structure. Thus the internal logic provides not just a language to describe properties, but also a way to identify and collapse unobservable differences.

Quantifiers as inference across morphisms: If $F: X \rightarrow Y$ is a morphism (a coarse-graining), then $\exists_F$ and $\forall_F$ allow us to relate predicates on X and predicates on Y . In logical terms, F can be seen as a **context change** or a **forgetful functor** that forgets some detail of X to produce Y . The pullback F^* corresponds to substitution or invoking context: given a property ψ of Y , $F^*\psi$ is the

property of X that holds when ψ holds for the image of x . The left adjoint $\exists_F$ is like the **$\exists$ -elimination** or **projection** of a property along F : $(\exists_F \varphi)(y)$ says "there is an x in the fiber over y such that $\varphi(x)$." The right adjoint $\forall_F$ says "for all x in the fiber over y , $\varphi(x)$." These allow *quantifying over the fiber* – which in programming language semantics would be like quantifying over all concrete states that correspond to an abstract state y .

For example, suppose $\varphi(x)$ is a predicate on a fine model X , and $F: X \rightarrow Y$ identifies some states (so Y is a quotient/coarse model). Then $\exists_F \varphi$ on Y is the predicate that holds at y in Y to the degree that *some* x in that equivalence class had $\varphi(x)$. If Y merges many X -states, $\exists_F$ picks up if any state in that collection satisfied φ . Dually, $\forall_F \varphi(y)$ is true only if *every* x mapped to y satisfied φ ; it is sensitive to the worst-case in the fiber. These constructs are particularly useful for **abstract reasoning**: one can derive properties of the abstract state y from properties of the concrete states in its fiber.

Compositional reasoning: Given two composable morphisms $X \rightarrowtail Y \rightarrowtail Z$, one can ask how quantification along $X \rightarrow Z$ relates to quantification along each piece. It turns out that $\exists_{G \circ F} = \exists_G \circ \exists_F$ and $\forall_{G \circ F} = \forall_G \circ \forall_F$ (this follows from uniqueness of adjoints or can be verified directly: taking sup over $F^{-1}(G^{-1}(z))$ is the same as first sup over $F^{-1}(y)$ then sup over $G^{-1}(z)$). Thus one can perform existential quantification in steps, which aligns with the idea of first projecting from X to Y , then from Y to Z . The Beck-Chevalley condition discussed above ensures that if we have a diagram with a pullback, moving a predicate around the square yields the same result, reinforcing consistency when reasoning about a situation from different perspectives. For instance, consider a scenario: X is a detailed model, Z is another detail, and Y is a common abstraction of both (X and Z relate via a pullback). Then Beck-Chevalley says that if we want to compute $\exists$ of some φ along two routes (via X then to Y , or first pull X and Z together then $\exists$ on Z), we get the same answer ²⁹. This is a form of **parametricity** or **independence of irrelevant detail**: the truth of an existential statement does not depend on which path of abstraction we take, as long as the diagram commutes.

The Frobenius reciprocity laws further allow moving quantifiers across known facts (ψ) that do not depend on the quantified variable. For example, $\exists_F(\varphi \wedge F^*\psi) = (\exists_F \varphi) \wedge \psi$ ³⁹ means if we want to check "there exists an x in fiber of y such that $\varphi(x)$ and $\psi(y)$," we can equivalently check " $\psi(y)$ holds, and there exists such x satisfying φ ." If $\psi(y)$ is false, the whole statement is false; if $\psi(y)$ is true, it doesn't interfere with the existence of an x . Similarly for $\forall$, $\forall_F(\varphi \vee F^*\psi) = (\forall_F \varphi) \vee \psi$ says that if either the fiber has a property or ψ holds, it is the same as saying "for all x in the fiber property holds, or ψ holds." These properties mirror how one can take out conditions from under a quantifier if they are *outer* (depending only on the outside variable). They are important for proofs, allowing one to simplify quantified formulas.

In summary, the internal logic of $\mathbf{MMK}$ lets us talk about **stochastic predicates** on state spaces and how those predicates transform under **morphisms (coarse-grainings)**. A statement in this internal logic might look like:

$$Y \models \forall_F(\text{Safe}(x) \vee F^*(\text{Alarm}(y))),$$

meaning "for all x that map to y , either x is safe or there is an alarm at y ." By the Frobenius law, this is equivalent to $(\forall_F \text{Safe})(y) \vee \text{Alarm}(y)$ for each y , which is a more intelligible

condition: "either all refined states are safe, or an alarm is present." Such translations are invaluable for reasoning about systems at different abstraction levels: $\forall F \text{ Safe}$ is a predicate on Y characterizing when an abstract state is entirely composed of safe concrete states.

Connections to Classical Categories and Further Commentary

The $\mathbb{MMK}$ framework generalizes and unifies several classical category-theoretic structures:

- **Topological/Metric spaces (Top/Met):** If we ignore the measure μ and kernel K , focusing only on the metric (or topology) and continuous maps, we recover a category similar to $\mathbf{Met}$ or $\mathbf{Top}$. Indeed, each $\mathbb{MMK}$ object has an underlying metric space $(|X|, d)$ and each morphism F is (optionally Lipschitz or at least measurable) mapping of the underlying spaces. If we further require F to be continuous and K to be trivial (identity kernel), $\mathbb{MMK}$ morphisms reduce to *isometric* or structure-preserving maps of metric spaces, and objects reduce to metric spaces with a probability measure (sometimes called metric measure spaces). There are known categories like $\mathbf{Meas}$ (measure spaces) or metric spaces, but $\mathbb{MMK}$ enriches those with a kernel.
- **Measure-preserving systems (Prob/Meas):** If we instead ignore the metric d and kernel K , an $\mathbb{MMK}$ object becomes just a probability space $(|X|, \mu)$ (with some negligible topology) and an $\mathbb{MMK}$ morphism $F: X \rightarrow Y$ reduces to a *measure-preserving map* ($F_{\#}\mu_X = \mu_Y$). In fact, taking K to be the identity kernel (no dynamics, a degenerate Markov chain that stays at the same state), the condition $K(F(x), B) = K(x, F^{-1}(B))$ becomes $1_{F(x) \in B} = 1_{x \in F^{-1}(B)}$ (always true), so any measurable F that pushes μ forward correctly is a morphism. Thus the subcategory of $\mathbb{MMK}$ with identity kernels is essentially the category of probability spaces with measure-preserving maps ⁶. This is sometimes denoted $\mathbf{Prob}$ or $\mathbf{Meas}_1$. Our internal logic in that case collapses to the σ -algebra logic of events (predicates are $[0,1]$ -valued measurable functions, which include event indicators). $\exists F$ and $\forall F$ then have interpretations in terms of images and preimages of sets under F (though $\forall F$ of an event is 1 iff the preimage event contains the whole fiber up to null sets).
- **Stochastic/Kernel maps (Markov processes):** If we drop the requirement that morphisms are functions and allow more general Markov kernels as morphisms, we would get the well-known **Markov category** (also called the category of stochastic maps, often denoted $\mathbf{Stoch}$ or similar). In such a category, an object is a measurable space and a morphism from X to Y is a Markov kernel $X \rightarrow \mathcal{P}(Y)$ (a stochastic relation). Composition is done via integration (Kolmogorov's composition of kernels). Our category $\mathbb{MMK}$ is more restrictive: each object carries its own kernel (dynamics), and morphisms are *functional* (deterministic) maps that respect those kernels. In a sense, $\mathbb{MMK}$ is a category of *labeled Markov processes* with a distinguished initial distribution, and morphisms are simulations or lumpings. This is analogous to the relationship between deterministic functions and general stochastic maps. One could embed $\mathbb{MMK}$ into a Markov category by sending a morphism F to the kernel $(x, B) \mapsto 1_{F(x) \in B}$ (the delta kernel corresponding to the function). The advantage of $\mathbb{MMK}$'s approach is that by keeping morphisms deterministic, we can talk about their *internal logic* as we did. Markov categories are typically not cartesian (no simple product of general kernels), whereas $\mathbb{MMK}$ retains finite limits by restricting to deterministic morphisms.

- **Probabilistic automata / Coarse-graining:** The notion of morphism here coincides with the concept of a **homomorphism of Markov processes** or a **coarse-graining of stochastic systems** used in various fields (e.g., lumpable Markov chains, sufficient statistics). For instance, if (X, μ, K) and (Y, ν, K') are Markov chains, $F: X \rightarrow Y$ being an $\mathbf{MMK}$ morphism means precisely that F is a *lumping projection* of the chain X onto chain Y ⁴⁰. In probability theory, this is sometimes called a *Markov chain factor* or a *Markovian projection*: it ensures that the image process $Y_t = F(X_t)$ is itself a Markov chain with kernel K' , and that F maps the stationary (or initial) distribution of X to that of Y . Classic results in lumpability (like if a partition yields constant transition rates across equivalence classes, then it defines a lumpable quotient) are captured categorically by the existence of a morphism. So $\mathbf{MMK}$ can be seen as a category of probabilistic models and their *exact coarse-grainings*.
- **Comparison with $\mathbf{Top}$ and others:** While $\mathbf{MMK}$ is not a topos (since it lacks certain colimits and a subobject classifier for all subobjects, etc.), it parallels $\mathbf{Top}$ (topological spaces) in that each object has an internal logic (in $\mathbf{Top}$, an internal intuitionistic logic of open sets or closed sets can be considered). In $\mathbf{MMK}$, the internal logic is $[0,1]$ -valued and σ -complete, reflecting measure-theoretic truth rather than just open set membership. The presence of $\exists, \forall$ adjoints is akin to $\mathbf{Top}$ having quantifiers under certain conditions (like proper maps inducing existence of right adjoints on subobject lattices, etc.), but here quantifiers always exist due to the nature of taking sup/inf.
- **Embedding classical categories:** There are canonical functors from several standard categories into $\mathbf{MMK}$. For example, any **measurable space** X with a probability measure μ can be embedded as $(X, d_{\text{disc}}, \mu, \mathrm{id})$ where d_{disc} is a trivial discrete metric (or 0 metric distinguishing points), and id kernel (no randomness, stays at the state). Then morphisms between such objects are exactly measure-preserving maps. Similarly, any **deterministic dynamical system** can be seen as $\mathbf{MMK}$ object with $K(x, \cdot) = \delta_{T(x)}$ for some function $T: X \rightarrow X$, making it a degenerate Markov chain (this would require T to preserve measure if we want stationarity). Any **topological space** (Polish) with a Borel probability measure can be included by forgetting topology (except as support of measure) and adding trivial kernel. Interesting inclusions mentioned in KUSMAN framework include groups, rings, and Hilbert spaces ⁴¹ ⁴²: for instance, a group G with a word metric, Haar (or counting) measure, and a random walk kernel (such as simple random walk on the Cayley graph) is an $\mathbf{MMK}$ object ⁴³. Group homomorphisms that respect measure (e.g. finite-index subgroups mapping) can give $\mathbf{MMK}$ morphisms ⁴⁴. Hilbert spaces with measures (Gaussian or Plancherel) and kernels (quantum channels or diffusions) can also be embedded. These embeddings are **faithful** (they reflect distinct maps as distinct $\mathbf{MMK}$ morphisms) and often **strong monoidal** (preserving products), as well as preserving the Q functor (coarse-graining). Thus $\mathbf{MMK}$ serves as a unifying arena where various mathematical structures (analytic, algebraic, probabilistic) are seen as special cases of metric-measure systems with dynamics.
- **Reflective subcategories via closures:** As indicated earlier, the KUSMAN framework defines four closure operators $\mathrm{clos}\{a\}, \mathrm{clos}\{g\}, \mathrm{clos}\{s\}, \mathrm{clos}\{\ell\}$ on $\mathbf{MMK}$, each capturing a form of simplification or reduction ⁴⁵. Each $\mathrm{clos}\{x\}$ is an idempotent endofunctor that reflects X into a subcategory of “ x -closed” objects ⁴⁶ ⁴⁷. For example, $\mathrm{clos}\{a\}$ (algebraic closure) might produce the minimal automaton or abstract machine equivalent to X (like identifying states that are behaviorally equivalent), $\mathrm{clos}\{g\}$ (geometric closure) might enforce completeness or

compactness, $\text{clos}\{s\}$ (spectral closure) might collapse fast transitions or small scales (e.g. merging states that mix rapidly), and $\text{clos}\{\text{ell}\}$ (logical closure) we discussed as merging logically indistinguishable states ³⁶. These are all **reflective subcategories** because each $\text{clos}\{x\}$ has a unit $\eta^x: \text{Id} \rightarrow \text{clos}\{x\}$ that is universal (any map from X to an $\text{clos}\{x\}$ -closed object factors through η^x_X) ⁴⁷. The existence of finite limits in MMK and the adjoint functor theorem (in a suitably complete category) help guarantee these reflectors exist and are well-behaved. The internal logic plays a key role especially for $\text{clos}\{\text{ell}\}$: $\text{clos}\{\text{ell}\}(X)$ is essentially the *hereditary image* of X under all predicates in $\text{Pred}(X)$, ensuring that the logical structure is complete. Also, conditions like completeness or compactness in metric terms are ensured by $\text{clos}\{g\}$ or $\text{clos}\{s\}$ which might involve taking limits (which we know exist) or closures under certain operations. Each closure operator $\text{clos}\{x\}$ being extensive and idempotent means $\eta^x_X: X \rightarrow \text{clos}\{x\}(X)$ is a monomorphism embedding X densely or fully into a simplified version, and applying the closure again does nothing new ¹¹.

Finally, it's worth emphasizing how all these foundational properties facilitate higher-level results in the KUSMAN $\text{Q}-\text{varsigma}$ framework. The MMK category's laws ensure that **complex multi-layer constructions decompose nicely**. For example: - In the **Hodge layer** (referring perhaps to a scenario involving Hodge decomposition or continuous dynamics), the existence of products means one can combine a physical continuous system with a probabilistic one, and the internal logic might be used to enforce conditions like orthogonality or integrality via predicates. The formal quantifiers guarantee that constraints (like "for all time steps, energy is conserved") can be represented. - In the **Birch-Swinnerton-Dyer (BSD) layer** (which might relate to p -adic dynamics on elliptic curves, as hinted by closures in a number theory context ⁴⁸ ⁴⁹), the finite limits and internal logic allow us to talk about local-global principles. Pullbacks model combining local conditions, and $\exists, \forall$ can express existential solutions or universal constraints across fibers of a fibration (like considering all primes or all Selmer group elements). The reflection properties of closures ensure we can isolate the "closure" of a condition (like the Selmer group closure) as a subobject which is itself an MMK object ⁵⁰. - In the **Riemann Hypothesis (RH) layer** ⁵¹, one deals with an MMK object capturing spectral properties of the zeta function (as described in the closures snippet). The logical closure $\text{clos}\{\text{ell}\}$ might encapsulate enforcing all predicates (like positivity or symmetry conditions) needed for the RH to hold. The existence of adjoints and limits assures that the necessary logical constraints can be imposed categorically (e.g., taking an equalizer to impose functional equation symmetry, as a kind of sigma_{FE} closure in the snippet ⁵²).

All these advanced applications rely on the fundamental categorical infrastructure we have established: without a well-defined category with products and an internal logic, one could not rigorously talk about *combining systems, comparing them, or enforcing multi-level constraints*. The fact that MMK has symmetric monoidal structure means composite systems (like combining an algebraic system with a geometric one) reside in the same category and the closures can act on them componentwise. Finite limits guarantee that any finite diagram of interest (like an equalizer imposing an equivalence relation, or a pullback gluing two systems along a common factor) has a solution in MMK . The hyperdoctrine allows us to express and manipulate conditions on systems in a formal way, rather than ad-hoc.

Foundational Summary

To conclude, we enumerate the key foundational results proved above and note their roles in the broader Σ -Q (KUSMAN) framework:

- **Category Axioms:** We defined objects $X=(|X|,d,\mu,K)$ and morphisms as measure-preserving, kernel-intertwining maps. We showed $\mathbf{MMK}$ is a category (identities and associative composition exist) ⁹. *This ensures that complex constructions of systems via composition are well-defined, enabling multi-step coarse-grainings and functorial operations (like the Σ functor) to be applied sequentially without ambiguity.*
- **Finite Limits:** $\mathbf{MMK}$ has a terminal object (the unit state) and binary products (independent product of systems) ¹², as well as equalizers of parallel morphisms (subsystems where two descriptions coincide) ¹⁷ ⁷. Thus all finite limits exist. *These provide the ability to form new systems from old: the terminal object serves as an identity for monoidal structure and a target for state-property maps; products allow building joint systems and underpin the monoidal composition of closures and functors; equalizers allow imposing equivalences or constraints, which is crucial for constructing quotient systems like the Σ functor outputs and for defining subobjects that satisfy certain equations (e.g., fixed-point conditions in dynamical systems). Finite limits also guarantee the existence of pullback squares used in proving logical consistency (Beck–Chevalley), which is needed for patching together local conditions as done in number-theoretic applications (BSD layer) or in matching different perspectives of a system.*
- **Internal Predicate Functor:** We defined $\text{Pred}(X)$ as the complete lattice of $[0,1]$ -valued predicates on X ⁵³, with standard fuzzy logic operations, and F^* as pullback of predicates along F ²³. This provides a language to talk about properties of states and to compare states across different systems. It is the foundation for defining when one system simulates or approximates another in terms of property satisfaction (e.g., logical closure in the logical layer identifies states with identical predicate truth values ³⁶).*
- **Existential and Universal Quantifiers:** For each morphism $F:X \rightarrow Y$, we established left adjoint $\exists_F$ and right adjoint $\forall_F$ given by essential sup/inf over fibers ²⁵, with the adjunction laws ²⁷. *These quantifiers allow one to reason about one layer of abstraction in terms of another: $\exists_F$ and $\forall_F$ translate between local (fine) properties and global (coarse) properties. In practice, they enable the formalization of statements like "there exists a refined state with property P" or "for all refined states, property Q holds" directly in the logic of the coarse system. This capability is crucial in proving the correctness of coarse-graining (Σ) – for instance, showing that if a property is invariant under the closures, it can be regarded as a property of the quotient – and in relating different closure operators (one can formally show $\text{clos}\{\ell\}$ yields a quotient where for all distinct states some distinguishing predicate exists, etc.).*
- **Beck–Chevalley Condition:** In any pullback square, we have $\exists_{\pi_Z} \circ \pi_X^* = G^* \circ \exists_F$ and $\forall_{\pi_Z} \circ \pi_X^* = G^* \circ \forall_F$ ²⁹. *This condition ensures consistency of logical reasoning when composing quantifications with structural connections. It was and will be used to interchange steps like "restrict context then quantify" vs "quantify then restrict," which appear in proofs that certain diagrams commute or certain reflectors preserve structure. For example, in the BSD application, one might have a pullback relating local and global conditions; Beck–Chevalley guarantees that existentially aggregating local solutions and then comparing with global conditions is equivalent to first gluing local*

pieces (pullback) then existentially checking – a necessary fact for proving that local-global principles (like Selmer conditions) behave well under $\mathcal{Q}$. In the RH context, multiple constraints (like functional equation, positivity, etc.) come from different directions and one needs to ensure applying closures in any order (the order-independence of $\mathcal{Q}$ closures ⁵⁴ ⁵⁵) yields the same result, which conceptually relies on a Beck–Chevalley type interchange of operations.

- **Frobenius Reciprocity:** We proved $\exists_F(\varphi \wedge F^\wedge \psi) = (\exists_F \varphi) \wedge \psi$ and $\forall_F(\varphi \vee F^\wedge \psi) = (\forall_F \varphi) \vee \psi$ ³¹. These identities let us pull outside conditions in or out of quantifiers, simplifying complex predicate formulas. They play a role in showing that certain predicates are invariant under closure or that certain closures commute. For instance, to prove a property of the composite closure $\mathcal{Q}$ (which is a sequence of closures $\text{clos}\{x\}$), one often needs to move a condition across a quantifier associated with one closure to apply another – Frobenius laws justify such moves. They also underpin the fact that our internal logic is well-behaved (it is a first-order logic that respects conjunctions and disjunctions with quantifiers), which is needed to argue about completeness or soundness of the logic (ensuring no contradictions arise when combining quantifiers and connectives).
- **Symmetric Monoidal Structure:** $\mathcal{MMK}$ is symmetric monoidal under $\otimes$ with unit I . This allows treatment of independent compositions of systems. All four closure operators $\text{clos}\{a\}, \text{clos}\{g\}, \text{clos}\{s\}, \text{clos}\{\ell\}$ respect this monoidal structure (as they correspond to applying certain operations factorwise on independent systems) ⁵⁶. For example, geometric closure of X and Y individually implies geometric closure of $X \otimes Y$, etc. The monoidal structure also enables discussions of entanglement of logic – e.g., $\text{Pred}(X \otimes Y)$ relates to $\text{Pred}(X)$ and $\text{Pred}(Y)$ (in fact, $\text{Pred}(X \otimes Y)$ contains predicates that can often be factored or approximated by predicates in each component). This will be relevant in discussing interactions between algebraic and logical closures (where one might consider combined properties like “both algebraic and logical conditions hold,” which factorize over closures).

Each of the above foundational laws contributes to a robust **categorical toolkit** for $\mathcal{MMK}$. In later layers of the KUSMAN framework, these results are invoked frequently: for instance, proving that $\mathcal{Q}$ (the composite closure) is functorial and idempotent uses the universal properties of products and equalizers; showing that $\text{clos}\{\ell\}$ creates the *coarsest* quotient preserving all predicate truth requires the internal logic’s completeness and the existence of $\forall_F$ to enforce that preservation ³⁷; establishing that different closure operators commute (as seen with $\mathcal{Q}_p$ having order-independent composition ⁵⁷) leverages the interchange properties akin to Beck–Chevalley (since each closure may quantify or constrain along a different morphism, and one needs to swap the order).

In conclusion, the **Foundations of $\mathcal{MMK}$** we have presented provide a formal, rigorous basis for the advanced developments in the KUSMAN $\mathcal{Q}$ – varsigma Master Theorem. By solidifying $\mathcal{MMK}$ as a category with finite limits and an internal hyperdoctrine, we ensure that subsequent categorical constructions (reflective subcategories, functorial quotients, embeddings of other categories, etc.) stand on firm ground. These laws are the bedrock upon which the higher-level results (Hodge decomposition interpretations, p -adic closure properties for BSD, and logical invariants for RH) are built, guaranteeing consistency and coherence in the unifying framework. The careful interplay of metric, measure, and kernel aspects through category theory and logic is what allows KUSMAN’s ambitious synthesis of disparate mathematical phenomena into a single structure to succeed.

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